Colby Baker

Software Engineer | Backend APIs and LLM systems | Java, Python, TypeScript
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Experience

CVS Health

Software Engineer, Health Agent AI Team *May 2026 – Present*

Patient-facing AI health assistant in CVS Health's consumer app. Python, LangGraph, FastAPI, React, TypeScript, Splunk, Kubernetes.

- Built a session-debugging platform with **React and FastAPI** that pulls the raw Splunk logs for a patient session from every service that touched it (Scheduling, Pharmacy, Health Records, Health Counseling, Plan Benefits and the app shell), joins them into a **turn-by-turn timeline** and shows each turn's inputs, LLM output and downstream calls, replacing hand-written Splunk searches across **8 services**.
- Added failure attribution that names which service caused a bad turn and cites the log line as evidence, predicts what the patient actually saw by replaying the card-rendering rules, a per-patient view of every session, and an **analytics dashboard** (turn and per-node latency, retries, usage) built as cross-filtering views over in-memory fact tables, parity-checked against the team's Splunk dashboard.
- Made it fast and dependable: batch prefetching and caching cut cold loads from **~55 s to ~25 s**, bounded batch-analysis memory to stop 503s on large profiles, added a self-healing SSO transport for Splunk and a live query verifier, and built decryption for the field-level PHI log encryption I designed and rolled out to production. **30–40% of the team** adopted it on their own.
- Built a **~32k-line offline evaluation platform** for the assistant's answers (three-arm prompt comparisons, six-domain graders, blinded calibration, corpus replay) that consolidated 45 brittle checks into 5 judgment rubrics and gated the specialist rollout: two lab-answer defects fell from 25% and 43% of responses to zero, and blinded reviewers preferred the new answers **64%** of the time ($p = 0.0083$).
- Redesigned the monolithic health-summary step into **7 feature-flagged domain specialists** (vitals, labs, visits, medications and more) running in parallel with observable routing decisions, so each domain could be tuned, evaluated and rolled out on its own; piloted across five domains in staging.
- Shipped the recent-health-summary feature end to end across the **LangGraph** agent and the TypeScript card service (five-domain parallel fetch, LLM synthesizer, next-step chips; 14 merged PRs); raised lab-question recall from **84.3% to 92.1%** on 554 real lab panels with a fuzzy matcher; and cut the welcome turn by **~16.5 s** by replacing three sequential LLM calls with a deterministic route.

Software Engineer, API Scheduling Team *Jun 2024 – May 2026*

Appointment scheduling APIs for CVS Health's in-person and virtual care. Java, Spring Boot, GraphQL, Cassandra, MySQL, Kubernetes.

- Led the structured-logging and tracing modernization across the **3 core scheduling services** so one patient request can be followed end to end through asynchronous calls, using shared correlation IDs, parent spans and W3C trace context with **OpenTelemetry**; a 16.5k-event audit found and fixed lost correlation on **1.6%** of outbound calls.
- Owned releases for **19 months** across dev, UAT and two production regions in Harness, and built automation that opens promotion PRs from CI image tags, which shipped an **11-service UAT release wave in one day**; migrated **10 codebases** to GitHub Actions.
- Cut the scheduling API's full test suite by **58% (5m33s → 2m22s)** and one test group by **97% (185 s → 6 s)** by migrating to WireMock 3 and removing per-test Spring context rebuilds and test-state leaks.
- Built scheduling features end to end in **Java and Spring Boot**: earliest-availability and distance ranking for employer-sponsored provider search, and preferred-pharmacy capture carried across **6 services** through booking, confirmation and rescheduling.
- Built an LLM triage bot for lower-environment tickets with a newly hired engineer I mentored: it maps **44 GraphQL queries, 22 mutations and 60+ downstream APIs** into a knowledge graph, routes log traces through a gatekeeper model to **8 domain-expert nodes** and posts the root cause and owning team to Slack.

Digital Engineering Intern *May – Aug 2023*

- Revamped a leadership metrics dashboard with **React, TypeScript, Chakra UI and MySQL** (filters, quick-action toggles, data downloads) so leadership could pull their own reports instead of requesting them from developers.

Projects

Aspicio · aspicio.app *Next.js, TypeScript, Three.js, Supabase*

Astrology platform built solo: interactive birth charts, a 3D sky in Three.js and relocation maps on MapLibre, driven by an ephemeris engine I wrote on **NASA JPL DE441** (IAU 2006 precession, IAU 2000A nutation) whose positions match JPL Horizons within 0.0005" across a 1,250-cell test grid. **1,700+ commits** and 600 test files since July 2026.

Spotify Classic Lyrics · github.com/Colbyco *Electron, Node.js, GitHub Actions*

Desktop app that restores Spotify's lyrics panel on Windows and macOS, with undo, diagnostics and auto-update (4 releases).

Technical Skills

Languages and frameworks: Java, Python, TypeScript, SQL, Spring Boot, FastAPI, React, Next.js, Node.js, LangGraph, REST, GraphQL
AI systems: LLM applications, multi-agent routing, offline evaluation (graders, blinded review), guardrails, prompt design
Infrastructure and data: Kubernetes, Docker, GitHub Actions, Harness, Argo CD, PostgreSQL, MySQL, Cassandra, Splunk, OpenTelemetry

Education

Florida State University · B.A. Computer Science *2024*